

# Rigid transformations



1

- a Translation      b Rotation      c Reflection

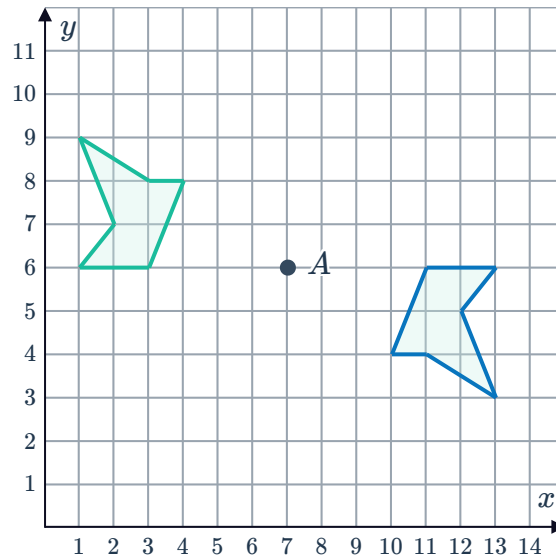
2

- a 2 units up      b 2 units right and 1 unit up  
c 4 units up      d 7 units down  
e 5 units left and 3 units down      f 4 units right and 2 units up  
g 2 units down      h 1 unit left and 1 unit down

3

- a Yes      b Yes      c Yes      d No  
e No

4



5

- a i  $270^\circ$       ii  $90^\circ$       b i  $180^\circ$       ii  $180^\circ$

6

- a Translation      b A translation 3 units right and 2 units up.

7

- a Reflection      b A reflection across the  $y$ -axis.

8

- a Rotation  
b A rotation of  $270^\circ$  anticlockwise around the origin.



## Congruence

10  $\angle RQP$

11  $\angle BAC$

12  $\angle ACB$

13  $\angle QRP$

14

a i 8 ii 7 iii 9

b  $\triangle XYZ$  congruent to  $\triangle ABC$

15

a  $EF$  b  $\angle EFD$  c 14 cm d  $\angle BCA$

16  $x = 41^\circ$

17  $x = 7, y = 13$

18  $\angle EFD$

19  $DF$

## Bisectors

20

a Yes b Yes c No d Yes  
e No f No

21

a Scalene b i Yes ii No iii No  
c No d Not equal



22

a Scalene

c No

b

i No ii No iii Yes

d Not equal

23

a Isosceles

c Yes

b

i No ii Yes iii No

d Equal

24

a Isosceles

c Yes

b

i No ii Yes iii No

d Equal